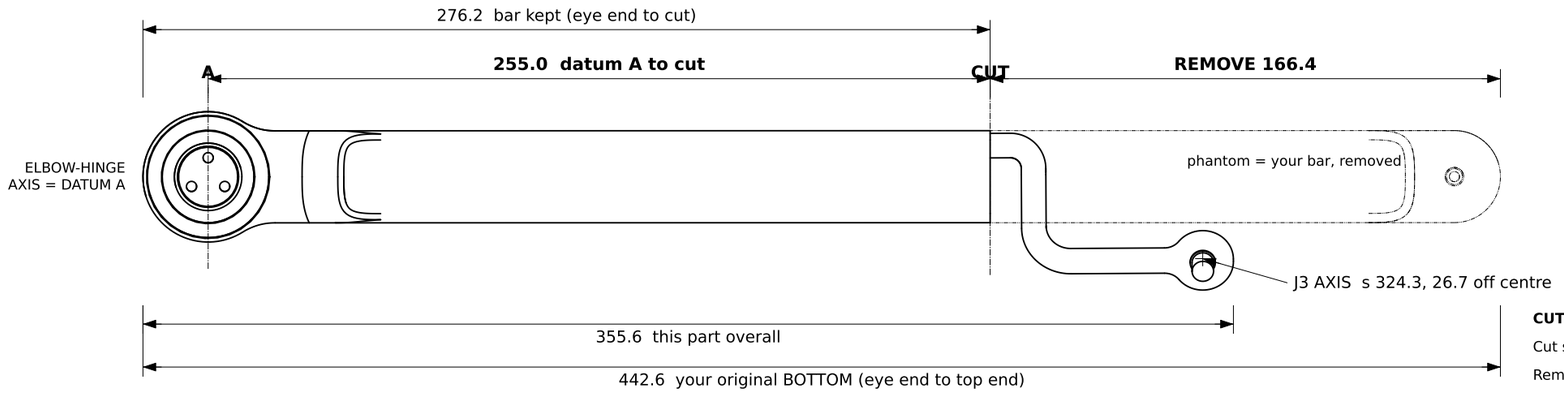
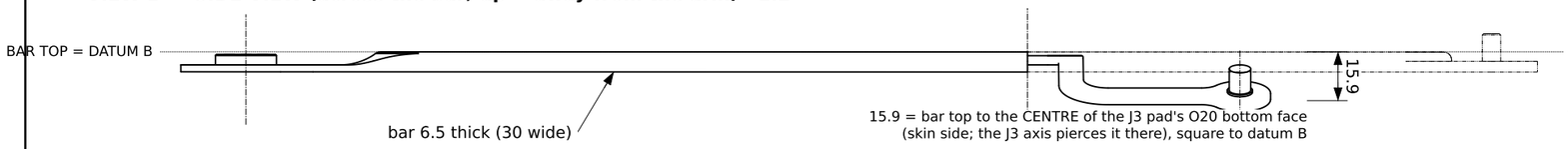
VIEW A - TOP VIEW ONTO THE BAR (normal to the bar top) 1:2



VIEW B - SIDE VIEW (across the bar; up = away from the arm) 1:2



CUT YOUR BOTTOM BAR

Cut square to the bar at s = 255.0 mm from the elbow-hinge axis (datum A).
Remove 166.4 mm from the top end: original eye-to-top 442.6 -> kept 276.2 mm.
Your cuff holes (3x2 M4 at s 94.25 / 124.25 / 154.25, v +-8.75) stay as they are.

ARM END (fused into the bar's channel, one part)

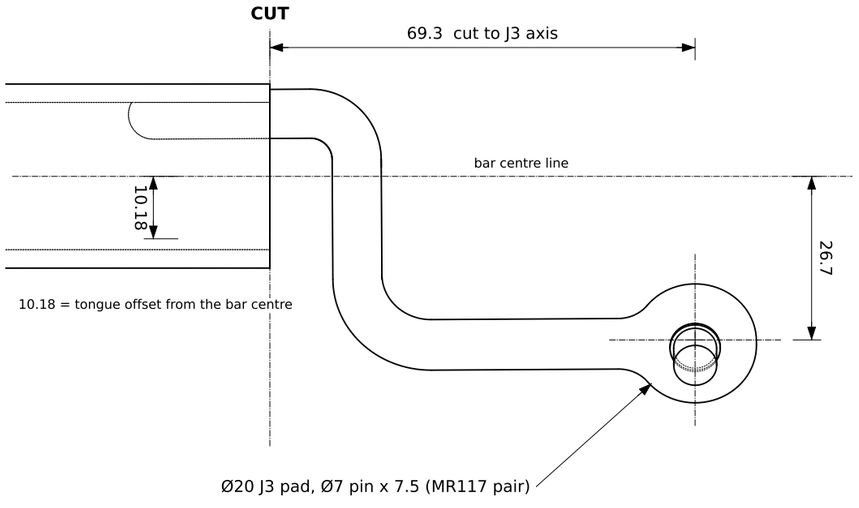
J3 axis passes through GH; pad face 71.85 mm from GH; at s 324.3, 26.7 off the bar centre,
On the J3 axis, square to the bar top (datum B): centre of the pad's bottom face 15.9, mid-plane 14.5,
pad top 13.1 below it (pad 3.0 thick). J3 axis tilted 21.8° from the bar normal toward -v (VIEW D).
Strap 8 x 3; pad plane vs channel plane 21.8° about a line 31.7 mm behind J3;
crossing 55 mm below J3; turns R12 (pad plane), R7.5 (channel); bend R4.5.
Tongue 10.18 mm off the channel centre, embedded 1 mm in the web.

UPPER-ARM CUFF (under the bar, for reference)

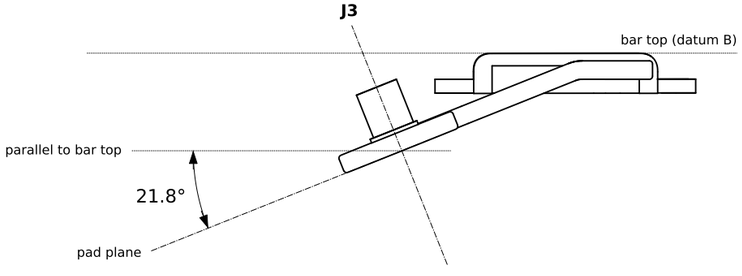
Straight extrusion s 83.85 -> 164.65 = 80.8 long; U: flat 70, side arcs R50,
wrap 110° each side, 2 mm wall, centred +4 mm across the bar; 2 mm liner inside.

Hidden lines omitted in A, B, D. Phantom = material removed from your bar.
All edges R0.5 unless shown. Projected from the built STEP solid.

DETAIL C - ARM END, TOP VIEW 1:1



VIEW D - END VIEW, DOWN THE BAR 1:1



BOTTOM (cut) + ARM END D6-04	
MATERIAL 7075-T6 (one 5-axis part), 79 g	SCALE 1:2 (A, B) 1:1 (C, D) UNITS mm, deg
exo-shoulder user-style 3R (r20 s1414 #4)	THIRD ANGLE DATE 2026-10-16 REV A